

* Desirable judgement

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Part A : short notes

1. Inferential Statistics :-
→ Using a sample to draw conclusions about a whole population.
2. Hypothesis Testing :-
→ A method to check a claim using data.
Parts : H_0 (no effect), H_1 (effect exists), α (significant level), test statistics, p-value, decision.
3. Confidence interval & Critical Value :-
→ A confidence interval is a range likely to contain the true value. The critical value is the cutoff that separates "reject" from "do not reject". eg:- ± 1.96 for Z at 5%.
4. P-Value :-
→ Probability of getting a result this extreme if H_0 is true. Small p-value means evidence against H_0 .
5. Type I vs Type II error :-
→ Type I = rejecting a true H_0 (false alarm).
Type II = failing to reject a false H_0 (missed effect).

6. Tests :-

→ Z-test : Compares means for large samples.

T-test : Compares means for small samples.

chi-square tests : tests the link between two categories.

ANNOVA : Compares means of 3 or more groups.

7. Covariance :-

→ Shows whether two variables move together (positive) or oppositely (negative).
Depends on units.

8. Correlation :-

→ Covariance scaled between -1 and +1.
Shows the strength and direction of a linear relationship.